MILITARY SPECIFICATION

CARBIDOPA AND LEVODOPA TABLETS

This specification is approved for use by the Defense Personnel Support Center, Defense Logistics Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Carbidopa and Levodopa Tablets.

1.2 Classification. The tablets covered by this specification shall be of the following strengths (see 6.2):

Strength 1 - 10 mg of carbidopa and 100 mg of levodopa
Strength 2 - 25 mg of carbidopa and 250 mg of levodopa

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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STANDARDS

FED-STD-140 - Tablets (for Medicinal Purposes).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Carbidopa and Levodopa Tablets, and shall be in accordance with all applicable requirements of FED-STD-140, and as specified herein.

S2. Classification. Shall be Type I.

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S5.3 The following is added to this paragraph:

Strength 1 shall be tablets containing 10 mg of carbidopa and 100 mg of levodopa per tablet, within the designated assay limits for the tablets.

Strength 2 shall be tablets containing 25 mg of carbidopa and 250 mg of levodopa per tablet, within the designated assay limits for the tablets.

The tablets shall assay to contain not less than 90.0 percent, and not more than 110.0 percent of the required amounts of carbidopa and levodopa when assayed as specified in 4.5.1.1 and 4.5.1.2.

3.2 Ingredients.

3.2.1 Levodopa. The levodopa used in the manufacture of the tablets shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.2.2 Carbidopa. The carbidopa used in the manufacture of the tablets shall be levorotatory and shall comply with the following requirements.

3.2.2.1 Identification. Shall comply with the identification test specified in 4.5.2.1.

3.2.2.2 Loss on drying. When dried to constant weight at 100°C in a vacuum oven at a pressure of less than 5 mm of mercury, the carbidopa sample shall lose not more than 7.9 percent of its weight using the USP method.

3.2.2.3 Residue on ignition. Shall be not more than 0.1 percent when determined using the USP method.

3.2.2.4 Heavy metals. Shall contain not more than 10 ppm heavy metals when determined by the USP procedure (Method II).

3.2.2.5 Assay. Shall contain not less than 99.0 percent and not more than 101.0 percent carbidopa (as monohydrate) when assayed as specified in 4.5.2.2.

3.2.2.6 Specific rotation. The specific rotation of the levorotatory carbidopa (as monohydrate) at 25°C shall be -21.0° to -23.5° when tested as specified in 4.5.2.3. at 589 mμ.

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3.3 Other ingredients. All other ingredients used in the manufacture of the tablets shall comply with S5.1 of FED-STD-140.

3.4 Expiration dating period. The expiration dating period shall be 60 months.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of the inspection requirements specified herein.

4.2 Sampling. The sampling as shown in FED-STD-140 shall be applicable.

4.3 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

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4.5 Tests.

4.5.1 Assay of finished tablets.

4.5.1.1 Carbidopa.

4.5.1.1.1 Reagents.

a. Sulfuric Acid (2N) - Dilute 60 ml of conc. H_2SO_4 to 1000 ml with purified water.

b. Sulfuric Acid (0.1N) - Dilute 3 ml of conc. H_2SO_4 to 1000 ml with purified water.

c. 2 Percent p-dimethylaminobenzaldehyde - Dilute 2 grams of p-dimethylaminobenzaldehyde to 100 ml with 2N H_2SO_4.

4.5.1.1.2 Standard preparation.

Accurately weigh 110 mg of hydrous carbidopa standard and transfer to a 100 ml volumetric flask. Dilute to volume with 0.1N H_2SO_4. Dilute 5 ml to 50 ml with purified water.

4.5.1.1.3 Sample preparation.

Weigh and finely powder 20 tablets. Accurately weigh a portion of the powder equivalent to about 25 mg of carbidopa and transfer to a 50 ml glass-stoppered centrifuge tube. Pipet 25 ml of 0.1N H_2SO_4 into the tube, stopper and shake for about 10 minutes, then centrifuge to obtain a clear supernatant. Dilute 5 ml of the clear solution to 50 ml with purified water.

4.5.1.1.4 Procedure.

Pipet 4 ml of the standard and sample solutions into separate 100 ml volumetric flasks, add 5 ml of p-dimethylaminobenzaldehyde and mix well. Heat the flasks at 100 - 105°C for 25 minutes. Cool the flasks in a water bath for 10 minutes. Dilute the flasks to volume with purified water and allow to stand for 20 minutes. Determine the absorbance of the standard and sample solutions at 453 nm using a suitable spectrophotometer and 0.1N H_2SO_4 in the reference cell.

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Calculation

$$\frac{Au}{As} \times \frac{Ws \times P}{} \times \frac{Wt}{Wu} \times \frac{Du}{Ds} = \text{mg anhydrous carbidopa per tablet}$$

Au = Absorbance of sample
As = Absorbance of standard
Ws = Weight of standard in mg
P = Purity of standard (expressed as anhydrous decimal fraction)
Wt = Average weight of 1 tablet in grams
Wu = Weight of sample in grams
Du = Sample dilution factor = $25 \times \frac{50}{5} = 250$
Ds = Standard dilution factor = $100 \times \frac{50}{5} = 1000$

4.5.1.2 Levodopa.

4.5.1.2.1 Materials.

a. TLC Plates - Silica Gel GF-250 microns 20 x 20 cm (Analtech Inc. or equivalent)

b. Chamber - Standard chromatographic developing tank that will accommodate 20 x 20 cm plates

c. Syringe - A 250 microliter #725 syringe

d. Visualization - UV Light

4.5.1.2.2 Reagents.

a. Sulfuric Acid (0.1N)

b. Hydrochloric Acid (0.05N)

c. Methanol

d. Developing Solvent I - Prepare a 200 ml mixture consisting of 40 ml each of benzene 1/, acetone, purified water n-butanol and glacial acetic aci

e. Developing Solvent II - Prepare a 200 ml mixture consisting of 80 ml methanol, 80 ml glacial acetic acid and 40 ml of chloroform.

1/ Other suitable solvent may be substituted for the benzene.

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4.5.1.2.3 Standard preparation.

Weigh accurately 125 mg of levodopa standard and transfer to a 50 ml volumetric flask. Add 25 ml of 0.05N HCl and shake. Dilute to volume with methanol and mix well.

4.5.1.2.4 Sample preparation.

Weigh and finely powder 20 tablets. Accurately weigh a portion of the powder equivalent to 125 mg of levodopa and transfer to a 50 ml volumetric flask. Add 25 ml of 0.05N HCl and shake for 15 minutes. Dilute to volume with methanol, mix well and centrifuge.

4.5.1.2.5 Procedure.

Score 2 parallel sides of the plate 1 cm from the edge. Score a third side 2 cm from the edge (this is the top). Then, score the plate into 3 equal sections between the parallel lines. Pour 200 ml of developing solvents into respective tanks and allow to equilibrate. "Streak" 200 microliters of the standard and sample preparation onto separate portions of the plate. An air jet may be used to aid in drying the streaks between applications. Place the plate in the chamber containing Solvent I and develop the plate to the top score. Remove the plate and air dry. Examine the plate under shortwave UV light and mark the area(s) containing Carbidopa ($R_f \cong 0.82$). Remove the Carbidopa band(s) by scraping with a scalpel and discard. Place the plate in the chamber containing Solvent II and allow to develop to the score mark formed by removal of the Carbidopa band. Remove the plate and air dry. Examine the plate under shortwave UV light and mark the areas containing levodopa ($R_f \cong 0.50$). Using a scalpel, remove the standard and sample bands onto glassine paper and transfer to separate 25 ml glass-stoppered centrifuge tubes. Pipet 10 ml of 0.1N H_2SO_4 into each tube, shake for 2 minutes and centrifuge to obtain a clear solution. Using a suitable spectrophotometer and 1 cm cells, scan the standard and sample solutions from 350 to 250 nm using 0.1N H_2SO_4 in the reference cell. Determine the absorbance at 280 nm and 310 nm.

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Calculation

$$\frac{Au - Bu}{As - Bs} \times \frac{Ws \times P}{} \times \frac{Wt}{Wu} = \text{mg levodopa per tablet}$$

Au = Absorbance of sample at 280 nm
As = Absorbance of standard at 280 nm
Bu = Absorbance of sample at 310 nm
Bs = Absorbance of standard at 310 nm
Ws = Weight of standard in mg
P = Purity of standard (expressed as a decimal fraction)
Wt = Average weight of one tablet in grams
Wu = Weight of sample in grams

4.5.2 Tests for ingredients.

4.5.2.1 Identification of carbidopa.

Disperse 50 mg of sample carbidopa in 10 drops of mineral oil. Press the dispersion between two sodium chloride plates. Record the infrared spectrum of the dispersion using a suitable infrared spectrophotometer. The spectrum obtained shall agree qualitatively with that of a carbidopa reference material similarly prepared and measured in the same spectrophotometer.

4.5.2.2 Assay for carbidopa.

Accurately weigh about 400 mg of carbidopa into a 250 ml Erlenmeyer flask and dissolve in 100 ml of glacial acetic acid. Titrate potentiometrically with 0.1N perchloric acid solution 1/ using a suitable platinum ring electrode as the indicating electrode and a suitable calomel electrode which has been emptied, rendered anhydrous by rinsing with glacial acetic acid and refilled with 0.1N lithium perchlorate in acetic anhydride. The end point of the titration is the point at which the greatest change in potential occurs per unit volume of titrant added.

1/ Preparation and standardization of this solution shall be in accordance with USP procedure.

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Calculation

$$\text{Percent Carbidopa (as Monohydrate)} = \frac{(Vx)\ (N)\ (244.24)\ (100)\ (92.6)}{(W)\ (100 - \text{percent L.O.D.})}$$

Where: Vx = Sample titer in ml
N = Exact normality of perchloric acid
W = Sample weight in mg
L.O.D. = Loss on Drying

4.5.2.3 Specific rotation.

4.5.2.3.1 Reagents.

a. Sodium hydroxide (0.25N) - Dissolve 10 gram of sodium hydroxide reagent in distilled water and dilute to 1 liter with distilled water.

b. Aluminum chloride reagent - Dissolve 65.0 gram of aluminum chloride ($AlCl_3 \cdot 6H_2O$) reagent grade in distilled water and dilute to 100 ml with distilled water.

c. Working solvent - To the aluminum chloride reagent solution add 0.25N NaOH with constant stirring until the pH approaches 1.5, then add drop-wise until pH 1.5 is obtained using a pH meter equipped with a calomel-glass electrode system. Neutralization to pH 1.5 normally requires from 60 to 70 ml of 0.25N NaOH.

4.5.2.3.2 Procedure.

Accurately weigh a 100 mg sample into a 10 ml volumetric flask. Dissolve in and dilute to the mark with working solvent.

Using a suitable polarimeter, determine the rotation of the sample in a one decimeter cell at 589 nm at 25°C.

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Calculation

$$[\alpha]_{589\ nm}^{25^{\circ}} = \frac{(a)\ (10)\ (92.6)}{(W)\ (100 - \text{percent L.O.D.})}$$

Where: a = measured rotation in degrees
W = sample weight in gram
L.O.D. = Loss on Drying

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 100 tablets of strength 1 or strength 2, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available bottle considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit packaging. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate packaging. Twelve (12) units of issue of strength 1 or strength 2 shall be packaged as specified in 5.2.7.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. One hundred forty-four (144) units of issue of strength 1 or strength 2 shall be packed as specified in 5.4.4 of PPP-C-186.

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5.3.2 Level B. One hundred forty-four (144) units of issue of strength 1 or strength 2 shall be packed as specified in 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number need not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. In addition, Type I shelf life marking as specified in MIL-STD-129 shall be shown.

5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following items:

National Stock Number	Item Identification	Specification Designation
6505-01-020-8279	CARBIDOPA AND LEVODOPA TABLETS, 100's	Strength 2
6505-01-020-8280	CARBIDOPA AND LEVODOPA TABLETS, 100's	Strength 1

Preparing activity:
DLA-DM

Project No. 6505-1790

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-C-37648(DLA-DM) CARBIDOPA AND LEVODOPA TABLETS

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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